

∴ In 5 days, working alone he can finish
 $5 \times 4 = 20\%$ of the work.

B can finish the work in 16 days.

So, in one day he can finish

$$\left(\frac{100}{16}\right)\% = 6.25\% \text{ of the work.}$$

Now, remaining work = $(100 - 20)\% = 80\%$

A and B together can do = $(4\% + 6.25\%)$

= 10.25%

Hence, time taken days

$$= \left(\frac{80}{10.25}\right) = 7.8 \approx 8 \text{ days (approx).}$$

So, total time taken = $5 + 8 = 13$ days.

49. (c) Area of field = πr^2

$$= \frac{22}{7} \times 42 \times 42 = 5544 \text{ ft}^2$$

$$\text{Number of days required} = \frac{5544}{154}$$

= 36 days

50. (b) Let the marked price of the machine be ₹100x.

S.P after a discount of 10% = $(100 - 10)\%$ of $100x = 90x$

$$\text{S.P after gain } 10\% = \frac{90x \times 110}{100} = 99x$$

If 99 is the S.P the marked

$$\text{price} = \frac{100x}{99x} \times 5940 = ₹6000$$

Amount of discount received by the retailer
 10% of 6000

$$= \frac{6000 \times 10}{100} = ₹600$$

51. (b) Indian Army's 'Major' rank is equivalent to 'Squadron Leader' of Indian Air Force.

This equivalence is part of the structured ranking system across the Indian Armed Forces, which ensures that ranks reflect similar levels of responsibility and authority across different branches

52. (c) **Lord Dalhousie:** Known for the Doctrine of Lapse.

Lord Ripon: Known for the Introduction of local self-government.

Lord Curzon: Known for the Partition of Bengal.

Lord William Bentinck: Known for the Abolition of sati system.

53. (c) A **stellar day**, also known as a **sidereal day**, is the **time it takes for the Earth to complete one full rotation on its axis relative to distant stars.**

It is approximately **23 hours, 56 minutes, and 4 seconds.**

This period is slightly shorter than a solar day, which is about 24 hours, because the Earth is also moving along its orbit around the Sun, requiring a little extra rotation to face the Sun again.

54. (b) The 17th edition of Joint Military Exercise SURYA KIRAN was conducted in Pithoragarh, Uttarakhand, from November 24th to December 7th, 2023.

It is a joint military exercise between the **Indian army and the Nepali army.**

It is an annual event and is conducted alternately in the two countries.

55. (c) Naval Detachment Minicoy has commissioned as **INS Jatayu**, an upgraded naval base by the **Indian Navy** to resolve to incrementally augment security infrastructure at the strategic Lakshadweep Islands.

INS Jatayu is the country's second naval base in Lakshadweep.

It will **enhance the operational reach of the Navy in the Indian Ocean**, a vital strategic imperative as China seeks to increase its footprint in the region.

The **Navy's first base on the islands, INS Dweep prakshak in Kavaratti, was commissioned in 2012.**

56. (d) **K P Sharma Oli was appointed Nepal's Prime Minister** for the third time on to lead the new coalition government.

K P Sharma Oli, 72, succeeds Pushpa Kamal Dahal Prachanda **who lost the vote of confidence in the House of Representatives**, leading to the formation of the new government process as per Article 76 (2) of the Constitution.

President Ram Chandra Paudel appointed **K P Sharma Oli as the new Prime Minister of the Communist Party of Nepal-Unified Marxist Leninist (CPN-UML)-Nepali Congress (NC) coalition.**

57. (a) **Nalanda University** was founded by the **Gupta Dynasty in the 5th century CE** during the reign of **Kumaragupta I.**

The university flourished under the patronage of the Gupta Empire in the **5th and 6th centuries CE.**

It attracted scholars and students from near and far, including **Tibet, China, Korea, and Central Asia.**

The Gupta Dynasty is known for its **cultural achievements and the liberal traditions** that resulted in a period of growth and prosperity for Nalanda until the 9th century CE.

58. (b) The Dead Sea has an **extremely high salinity level**, approximately **240 parts per thousand (ppt)**, making it one of the saltiest bodies of water in the world.

Red Sea: While it has high salinity, it is not as high as that of the Dead Sea.

Mediterranean Sea: The Mediterranean Sea has a salinity of about **38 ppt**, which is significantly lower than that of the Dead Sea.

Black Sea: The Black Sea has a much lower salinity, averaging around **15-18 ppt**, due to the influx of freshwater from rivers.

59. (b) The **Nagi and Nakti Bird sanctuaries situated in Jamui district in Bihar** are human-made wetlands and provide a natural habitat for a wide variety of flora and fauna, especially birds.

The sanctuaries are spread across 791 and 333 hectares and were developed primarily for irrigation through the construction of Nakti Dam.

60. (c) According to the recent Supreme Court directive, the percentage of posts in the Supreme Court Bar Association (SCBA) that must be reserved for women is **33%.**

The **Supreme Court mandated** that at least **one-third** of the posts in the executive committee of the SCBA should be reserved for women, which translates to a **minimum of 33%.**

61. (d) The **Namami Gange Programme** is implemented under the **Ministry of Jal Shakti**, specifically through the Department of Water Resources, River Development, and Ganga Rejuvenation.

This integrated conservation mission was launched to address pollution and promote the rejuvenation of the Ganga River.

62. (b) There are **5 players** in a basketball team.

It is played between two teams of **five players each on a rectangular court, usually indoors.**

Each team tries to score by tossing the ball through the opponent's goal, an elevated horizontal hoop and net called a basket.

63. (a) The **Indian Air Force's Rafale fighter jets** has participated in a military exercise in **France called Orion 23**, which involves thousands of **NATO** troops.

The exercise aims to enhance military cooperation and interoperability among participating nations.

64. (b) The sector that includes the industries where finished products are made from natural materials is known as the **Secondary Sector.**

The economic sectors are typically classified into three main categories:

Primary Sector: Involves the extraction and production of raw materials from the natural environment, such as agriculture, mining, forestry, and fishing.

Secondary Sector: Involves the processing and manufacturing of raw materials into finished goods. This sector includes industries like manufacturing, construction, and utilities.

Tertiary Sector: Involves the provision of services, such as retail, education, healthcare, and entertainment.

65. (d) A slam dunk, also simply known as **dunk**, is a type of basketball shot that is performed when a player jumps in the air, controls the ball above the horizontal plane of the rim, and scores by shoving the ball directly through the basket with one or both hands.

66. (d) **Arboriculture is the cultivation, management, and study of individual trees, shrubs, vines, and other perennial woody plants.**

It focuses on the care of trees and shrubs primarily for landscape and ornamental purposes.

Sericulture is related to the rearing of silkworms for silk production.

Horticulture involves the cultivation of fruits, vegetables, flowers, and ornamental plants.

Viticulture refers to the cultivation and harvesting of grapes, primarily for winemaking.

67. (a) Windmills are devices that convert kinetic energy (the energy of motion) from the wind into electrical energy.

Windmills do not directly convert solar, thermal, chemical, or nuclear energy into another form; they specifically harness the kinetic energy of the wind to produce electricity.

68. (c) **India successfully concluded hosting of the 46th Antarctic Treaty Consultative Meeting (ATCM-46) and the 26th Committee on Environmental Protection (CEP26) in Kochi, Kerala.**

The ATCM-46 was held with the overarching theme of *Vasudhaiva Kutumbakam*, a Sanskrit phrase that means one Earth, one family, one future.

69. (c) **"The God of Small Things"** was written by **Arundhati Roy**, an Indian author.

"The White Tiger" was written by **Aravind Adiga**, also an Indian author.

"Midnight's Children" was written by **Salman Rushdie**, an author of Indian descent.

"To Kill a Mockingbird" was written by **Harper Lee**, an American author, making it the book not written by an Indian author.

70. (c) **Statement 1 is correct: Every food web in every ecosystem is built on producers, who occupy what is known as the first trophic level of the food web.** Primary consumers, also known as herbivores, or animals that consume plants, make up the second trophic level.

Statement 2 is correct: Flow of nutrients in Ecosystem is cyclic. The biogeochemical cycle is the term used to describe the periodic flow of nutrients through an ecosystem. In an ecosystem, nutrients that are required for organism growth are continuously transferred from abiotic to biotic factors and from biotic to abiotic factors.

Hence both 1 and 2 statements are correct.

71. (b) **Bihar** was the first state in India to impose an **agricultural tax**.

This tax was introduced to generate revenue from agricultural produce, given that agriculture is a significant part of the state's economy.

72. (d) The Mumbai Trans Harbour Link, officially named as **Atal Bihari Vajpayee Sewri-Nhava Sheva Atal Setu** and colloquially known as **Atal Setu**, is a 21.8 km (13.5 mi) 6-lane grade separated expressway bridge, which connects **Mumbai with Navi Mumbai**, its satellite city.

73. (a) **Mission, Integrity, Excellence** is the core values of IAF, which state that the IAF is guided by these three core values that form the foundation of its operational philosophy and organizational culture.

74. (b) **INS Garuda** is a **naval air station of the Indian Navy**.

It is located in **Kochi, in the state of Kerala**.

It serves as a base for various naval aircraft and is one of the major air stations under the Southern Naval Command of the Indian Navy.

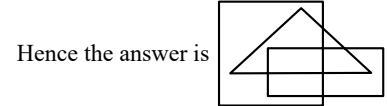
75. (c) **Sattriya Nritya** is a classical dance form that originated in **Assam, India**.

It was initially created as part of the **Sattras, the Vaishnavite monasteries**, as a cultural expression within the context of the **Bhaona performances**.

Sankardev, the 15th-16th century polymath from Assam, played a key role in developing **Sattriya Nritya** as a form of artistic expression within the region's broader cultural and religious practices.

76. (d) Here, in the given figure, first dot lies in square region only and second dot lies

in rectangle and square region and the third dot lies in the region common to triangle and rectangle. So, the option which consists of these regions is the answers.



77. (c)

A	B	C	D	E	F	G	H	I	J	K	L	M
Z	Y	X	W	V	U	T	S	R	Q	P	O	N

'SWISS' → opposite letters ← 'HDRHH'

'TABOO' → opposite letters ← 'GZYLL'

Similarly, 'REHAB' → opposite letters ← 'IVSZY'.

78. (b) "ADD" is written as "8"

Place value of A = 1, D = 4, D = 4

ADD → $(1 + 4 + 4) - 1 = 9 - 1 = 8$

Place value of S = 19, U = 21, B = 2

SUB → $(19 + 21 + 2) - 1 = 42 - 1 = 41$

Similarly,

Place value of M = 13, U = 21, L = 12

MUL → $(13 + 21 + 12) - 1 = 46 - 1 = 45$

79. (b) $(23 + 5) \times 2 = 56$

$= (6 + 7) \times 2 = 26$

$= (5 + 2) \times 2 = 14$

Similarly,

$= (7 + 5) \times 2 = 24$

80. (d) As in the coded language, Pink is called Brown. Therefore, Sky is Pink will be written as Sky is Brown.

81. (a) In the event of coincidence, the angle between the two needles will be zero degree.

$$\text{Angle} = \frac{60H - 11M}{2}$$

$$\therefore 0 = \frac{60 \times 2 - 11M}{2}$$

$$\Rightarrow 0 = 120 - 11M = 120$$

$$M = \frac{120}{11} = 10\frac{10}{11}$$

$\therefore$ The hands of the clock will coincide at $2 : 10\frac{10}{11}$.

82. (a) We know that in any two consecutive non-leap years, the exact date of the year will be one day ahead of the exact date of the previous year. If 4 December 2021 is on Saturday

Then, 4 December 2022 will be on: Saturday + 1 = Sunday.

83. (c) After interchanging position, Kaliya becomes 18th from left side.